Array 29 (2026) 100659

Contents lists available at ScienceDirect

Array

journal homepage: www.elsevier.com/locate/array

Trans-ADENet: Transformer-based Attention-guided Deep Ensemble Network for high-dimensional data classification

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ARTICLE INFO

Keywords:
High-dimensional data classification
Deep autoencoder
Transformer encoder
Self-attention mechanism
Ensemble learning

ABSTRACT

High-dimensional (HD) biomedical data, such as gene expression profiles and ECG signals, pose significant challenges for machine learning (ML) due to limited sample size, feature redundancy, and noisy distributions. Conventional models tend to overfit, whereas boosting and ensemble approaches struggle with irrelevant features. Deep autoencoders (DAE) reduce nonlinear dimensionality but miss complex dependencies, whereas transformers require large datasets to model long-range relationships through self-attention mechanisms. We propose a Transformer-based Attention-guided Deep Ensemble Network (Trans-ADENet) that integrates dimensionality reduction, attention-driven feature learning, and meta-level ensemble fusion in an end-to-end framework. A deep autoencoder compresses HD inputs into compact latent representations, refined by a Transformer Encoder with multi-head self-attention. Refined features are fed to diverse base classifiers (CatBoost, Support Vector Machine (SVM), TabNet, and Generalized Multi-Layer Perceptron (GMLP)), and their outputs are fused by a meta-MLP, which learns adaptive weights to yield robust predictions. Experiments on breast, leukemia, INCART2 and Thyroid-RNA datasets achieved 96.3%, 94.1%, 92.7% and 94.6% accuracy, surpassing state-of-the-art models in terms of accuracy, F1, precision, recall, and AUC. By combining representation learning, attention, and adaptive fusion, Trans-ADENet delivers accurate, interpretable classification for biomedical tasks.

1. Introduction

Significant advances in high-throughput data acquisition tools have led to the generation of high-dimensional, complex datasets in a wide range of scientific fields, including genomics, transcriptomics, proteomics, and biomedical signal processing [1,2]. Biomedical research and healthcare practices rely heavily on electrocardiogram (ECG) traces, high-dimensional omics profiles, gene expression arrays, and multi-sensor physiological recordings to understand disease development, treatment response, and biological system dynamics [3,4]. These high-dimensional regimes exacerbate the curse of dimensionality, in which the volume of a feature space increases exponentially with the number of dimensions, leading to sparsity and rendering distance-based learning metrics less informative. As a result, traditional machine learning algorithms overfit the training set, leading to poor generalization performance on new samples [5].

However, inherent heterogeneity, noise, missingness, and class imbalance in biomedical datasets pose challenges that require methodological advances to address and produce reliable predictions. Traditional machine learning algorithms, including logistic regression, k-nearest neighbours, and decision trees, cannot scale and generalize in high-dimensional settings [6]. Even ensemble methods, such as random forests and gradient boosting algorithms, suffer from performance degradation when overloaded with irrelevant features [7,8]. The deep learning models have proven to be the most effective tools for representation learning and big data analysis [9–11].

Transformer models, initially used for natural language processing, have used self-attention techniques to enhance deep learning performance [12,13]. It was introduced in 2017 and uses pure attention mechanisms rather than recurrent layers, thereby enhancing performance and training effectiveness across various tasks. This approach, combined with its self-attention mechanism, enables the model to understand the importance of each input sequence element, allowing it to encode complex patterns in the data, particularly crucial for high-dimensional classification tasks [14]. The effectiveness of transformers across various fields, including medical imaging and computer vision, is attributed to their encoder–decoder architecture and their ability to process long-range dependencies [15–17], a significant drawback of traditional convolutional neural networks [18]. However, the inductive bias inherent in convolutional networks, such as translation equivariance and locality, is often absent from standard transformers. This is

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https://doi.org/10.1016/j.array.2025.100659
Received 27 August 2025; Received in revised form 16 December 2025; Accepted 19 December 2025
Available online 30 December 2025
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a crucial component of their requirement to employ large datasets to attain reliable performance [19]. Autoencoder neural networks that learn to reconstruct their input provide a principled methodology for dimensionality reduction. By learning compressed latent representations that capture the underlying nature of the input data, autoencoders reduce the curse of dimensionality while retaining the features most useful for downstream tasks.

However, autoencoders may not be adequate for distinguishing complex interdependencies between features [20,21]. By applying Transformer Encoders to high-dimensional biological data, it is possible to assign importance weights to model features dynamically, enabling class discrimination and learning what to pay attention to by focusing on biomarker-related inputs. The application of transformer models to tabular, clinical, and omics data has opened new horizons for high-performance, understandable classification [22]. Multi-headed self-attention in nonlinear environments allows these models to reason about long-range feature interactions and perform sensitive feature selection during training. In addition to these developments, ensemble learning remains a significant approach to increasing the generalization and robustness of predictive models. By combining the outputs of multiple learners, ensemble methods reduce prediction variance, mitigate overfitting, and leverage the strengths of complementary inductive biases [23,24]. Multimodal fusion and adversarial robustness are introduced by the Robust Adversarial Fusion Transformer (RAFT), which aligns with the ensemble robustness discussion [25]. However, standard ensemble methods such as bagging and boosting assume static combinations of models and do not handle dynamic weights or higher-order feature interactions in predictions. This limitation has led to the adoption of meta-learning methods that learn to aggregate base-model outputs in a data-driven manner.

This work presents Trans-ADENet, a novel transformer-based attention-guided deep ensemble network for robust classification of high-dimensional biomedical data. The main contributions are summarized as follows:

- We design an integrated framework that combines deep autoencoding, transformer-based attention, and meta-level ensemble learning to tackle the challenges of high dimensionality, noise, and small sample sizes in biomedical datasets.
- A deep autoencoder module is employed to compress noisy, redundant input features into compact, information-rich latent representations, thereby alleviating the curse of dimensionality and improving downstream learning efficiency.
- The compressed latent features are further refined using a Transformer Encoder, which leverages multi-head self-attention to highlight globally informative and class-discriminative features dynamically.
- We incorporate a diverse ensemble of classifiers (Cat-Boost, Support Vector Machine (SVM), TabNet, Generalized Multi-Layer Perceptron (GMLP)) that capture complementary inductive biases, each tailored to distinct aspects of tabular, nonlinear, and structured data.
- A lightweight Meta-MLP is introduced to perform meta-level fusion of the base classifiers' outputs, learning dynamic, sample-specific weighting to enhance predictive robustness and generalization.

The rest of the paper is organized as follows: Section 2 This section reviews the related work. Section 3 provides a detailed description of the proposed framework. Section 4 presents experimental results and compares the performance of Trans-ADENet with baseline models. Section 5 concludes the paper with key conclusions and future research directions.

2. Related work

Recent advances in machine learning have led to a number of methods for handling high-dimensional biomedical data, which often suffer from noise, redundancy, and unbalanced class distributions. This section reviews key contributions in three main areas of our work: (1) dimensionality reduction and representation learning methods, especially deep autoencoders; (2) attention-based models, including transformer architectures for structured and tabular data; and (3) ensemble learning strategies with a focus on robust classification. These approaches form the foundation for developing the proposed Trans-ADENet model.

Deep learning models may automatically learn task-dependent features from raw input and perform effectively in numerous applications. In deep learning, feature fusion uses features from several layers or sources to create more informative and discriminative representations [26]. A key strength of deep learning is its ability to extract data-driven features for classification or regression tasks [27]. Deep learning models have demonstrated excellent performance in processing complex, high-dimensional, and noisy data [28]. High-dimensional eigenvalue problems are crucial in physics, chemistry, and engineering, but traditional numerical methods often struggle due to the curse of dimensionality. Neural network-based approaches, such as tensor networks(TNN), have demonstrated potential by facilitating the precise and efficient integration of high-dimensional data without the need for sampling [29,30].

Feature selection is a crucial step to identify useful features while minimizing computational complexity. Metaheuristic algorithms, such as Golden Jackal optimization (GJO), are widely used for efficiently exploring large search spaces. However, binary feature selection poses significant challenges, including premature convergence and an imbalance between exploration and exploitation. The improved Binary Golden Jackal optimization (BGJO) algorithm with multiple exploitation and exploration strategies was introduced in [31]. Multi-objective optimization approaches [32] are widely employed in feature selection problems, especially in high-dimensional datasets, to balance classification accuracy and feature subset compactness.

Transformer architecture revolutionizes sequence models by incorporating attention mechanisms, thereby enhancing sequence-to-sequence models in natural language processing (NLP) and other domains. According to [33], current attention mechanisms, including additive, multiplicative, and scale dot-product models, classify variants of attention in deep learning models. The Transformer models, such as DeepGene, which utilizes a Pan-Genome Graph Transformer to learn long-range DNA associations, demonstrate the growing importance of transformer models in omics sequence modelling [34]. Its capacity to model dependencies via self-attention has led to outstanding performance in visual, signal, and biomedical applications [35–38]. Recent foundation models, such as Mouse-Geneformer, demonstrate that transformer-based models can be applied in transcriptomics to learn cross-species generalization and how to train on massive biological datasets [39]. Transformers are promising approaches for image classification and multimedia applications, particularly for high-dimensional data. An improved transformer-based dehazing network that combines attention and frequency-domain methods was demonstrated in [40]. Hybrid architectures are increasingly being used in omics analyses. To improve single-cell annotation prediction and learn cell associations, scGraPhT [41] combines Transformer Encoders and graph neural networks.

Ensemble learning has overcome the limitations of individual classification algorithms by combining their strengths, thereby boosting weak classifiers and improving model stability [42,43]. Ensemble approaches have already been proven to enhance classification stability and accuracy, particularly for high-dimensional datasets [44]. Deep ensemble networks outperform other methods by reducing overfitting and improving prediction accuracy by leveraging the diversity of multiple deep learning models [45]. Attention-based deep ensemble networks

are introduced to help the model focus on the most important features among multiple classifiers [46].

Deep learning models have shown increasing utility in biomedical applications, particularly for analysing high-dimensional microarray data to classify diseases and discover biomarkers [47]. However, many existing methods fail to account for cross-study variability, such as batch effects, which can significantly impact model performance when integrating data from multiple sources. Variational autoencoders (VAEs) have emerged as an effective method for developing compact and interpretable latent representations of high-dimensional physiological signals [48]. Meanwhile, frameworks like VAE-FastGA subspace search model efficiently filter out anomalous subspaces from weakly correlated, ultra-high-dimensional data [49]. Similarly, interpretable fuzzy systems and evolving neural networks have been developed for feature compression in complex, high-dimensional environments [50,51]. Other hybrid approaches, such as the weighted Fisher score (WFISH) [52] and multi-objective particle swarm optimization [53], strike a balance between feature relevance and computational cost. Recent models, such as DeepGene Transformer, demonstrate their effectiveness by combining attention mechanisms with deep representation learning for cancer subtype classification and gene expression analysis [54]. DeepEnsemble Transformers(DETs), a framework for high-dimensional biomedical data that integrates ensemble learning, transformer-like multi-layer refinement, and deep neural networks. DETs use stacked ensemble layers (e.g., random forests, AdaBoost, gradient boosting) to progressively denoise and reshape features, followed by feed-forward layers that further refine the representation. These ensemble transformations act similarly to attention, emphasizing informative regions of the feature space [55]. A variational quantile classifier [56] was proposed for effective high-dimensional data classification.

The above literature demonstrates the growing importance of hybrid approaches that combine dimensionality reduction, attention approaches, and ensemble learning to improve classification performance on high-dimensional biomedical datasets. While individual methods offer specific strengths such as improved feature extraction or robustness to noise, only a few integrate these components into a unified, end-to-end architecture. Motivated by these gaps, our proposed model, Trans-ADENet, synthesizes deep autoencoding, transformer-based attention, and heterogeneous ensemble learning to address the main challenges of high-dimensional biomedical classification. The following section describes the architecture and methodology of Trans-ADENet.

3. Methodology

Fig. 1 illustrates the proposed Trans-ADENet model, which is carefully designed for addressing the inherent difficulties and bottlenecks of high-dimensional data classification. It operates by utilizing the synergistic interaction of three key computational modules: (1) nonlinear compression of features with a deep autoencoder, (2) context-aware refinement of features with transformer-based multi-head self-attention mechanisms, and (3) generalized and robust prediction with an ensemble of heterogeneous classifiers associated with a meta-level neural fusion approach. The proposed architecture represents a modular yet end-to-end trainable pipeline that effectively mitigates the curse of dimensionality and enhancing classification accuracy, generalization, and interpretability even in the presence of noise, redundancy, and small-sample scenarios.

First, the deep autoencoder compresses the high-dimensional input feature space into a compact latent representation. This reduction step removes irrelevant and redundant features while preserving the key structural patterns necessary for downstream applications. This unsupervised, encoder–decoder model learns low-dimensional manifolds that capture important data diversity and enhance feature decomposition.

Next, Transformer Encoder block refines the latent representations obtained from autoencoder. It uses multi-head self-attention to rescale and reweight feature embeddings based on their global contextual importance and higher-order interactions between model features. By focusing attention to the most informative components, this step helps to reduce overfitting and noise sensitivity.

Finally, refined feature reprasentations are passed as input to an ensemble of diverse classifiers — CatBoost, SVM, TabNet, and GMLP, each contributing unique biases and learning approaches suited for complex tabular data and probabilistic outputs are combined through a meta-MLP fusion layer in the meta decision stage. The feedforward neural network identifies latent relationships and inter-model dependencies, allowing classifiers to assign adaptive weights and rectify poor predictions. The adaptive stacking approach enhances the predictive performance and model robustness across diverse datasets.

Overall, Trans-ADENet combines dimensionality reduction, attention-driven feature refinement, and deep ensemble learning into a cohesive architecture. Its end-to-end optimization ensures all component co-adaptation during training, resulting in an accurate and resilient model for complex, high-dimensional classification tasks. Following sections provide a detailed technical discussion of its design, computational formulations, and training methods.

3.1. Data preprocessing and normalization

In biomedical applications, the input data often contains missing values, heterogeneous feature scales, and categorical labels, all of which must be carefully trans formed before model training. To improve the stability of the proposed Trans-ADENet model effective preprocessing is essential and it contains three steps: Handling missing values, Standardization and Label Encoding.

Handling missing values

To handle missing entries, a mean imputation strategy is employed. Specifically, for each feature, missing values are replaced by the mean of the observed (non-missing) values of that feature. Formally, if x_{ij} denotes the value of the jth feature for the ith sample, the imputed value $\tilde{x}_{ij}$ is defined as in Eq. (1).

$$\tilde{x}_{ij} = \begin{cases} x_{ij}, & \text{if } x_{ij} \text{ is observed} \\ \mu_j = \frac{1}{|O_j|} \sum_{i \in O_j} x_{ij}, & \text{if } x_{ij} \text{ is missing} \end{cases} \tag{1}$$

where O_j denotes the set of indices for samples where the jth feature is observed, and μ_j is the mean of the available values of that feature. This imputation preserves the overall distribution of the feature while enabling a complete data matrix for model training.

Standardization

After imputing missing data, the features are standardized to remove scale discrepancies that could bias the learning process and it centres each feature at the mean and scales it to unit variance, ensuring uniform contributions across all features. For the imputed data matrix $\tilde{X} \in \mathbb{R}^{n \times m}$ where n is the number of samples and m the number of features, the standardized feature z_{ij} is calculated by using Eq. (2).

$$z_{ij} = \frac{\tilde{x}_{ij} - \bar{x}_j}{s_j} \tag{2}$$

where

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^{n} \tilde{x}_{ij} \quad \text{and} \quad s_j = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (\tilde{x}_{ij} - \bar{x}_j)^2} \tag{3}$$

are the empirical mean and sample standard deviation of the jth feature across all samples. Through this transformation, each feature follows a standard normal distribution with a mean of 0 and a variance of 1, which is beneficial for the convergence of gradient-based optimization algorithms commonly used in deep learning.

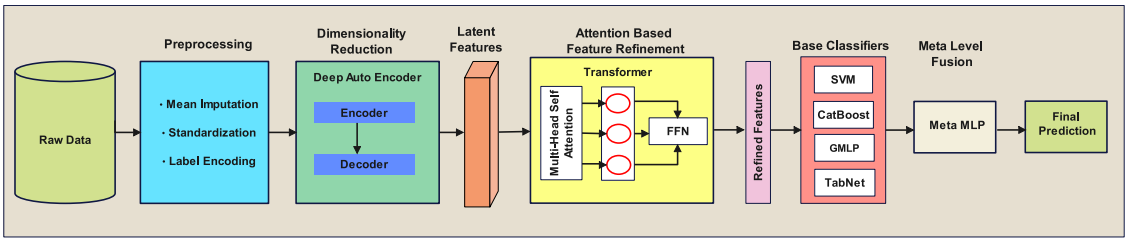


Fig. 1. Integrated Architecture of Trans-ADENet.

Label encoding

Finally, the categorical target labels must be converted into numerical form for model compatibility. This is achieved through label encoding, where each unique class label ck from the set of classes $C = c_1, c_2, \ldots, c_K$ is mapped to an integer index $k-1$. Formally, a mapping function is defined such that for each sample's label y_i, its encoded form is mentioned in Eqs. (4) and (5).

$$LE : C \rightarrow \{0, 1, \ldots, K-1\} \tag{4}$$

$$y_i^{\text{encoded}} = LE(y_i) \tag{5}$$

This encoding facilitates supervised learning models to treat the problem as a multi-class classification task. Collectively, these preprocessing steps transform the original dataset X and labels y into a fully imputed, standardized feature matrix Z and integer-encoded label vector $y^{encoded}$, thereby ensuring the data is in an optimal state for downstream representation learning and classification.

3.2. Dimensionality reduction with deep autoencoder

To address the high dimensionality and redundancy inherent in complex biological datasets, our framework leverages a DAE for nonlinear dimensionality reduction and noise filtering. The autoencoder consists of two key components: an encoder and a decoder, parameterized by θ and ϕ, respectively.

3.2.1. Designing of encoder and decoder

The encoder function is represented as in Eq. (6).

$$f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^k \tag{6}$$

where f_θ maps the original high-dimensional input vector $x_i \in \mathbb{R}^d$ into a compact latent representation as mentioned in Eq. (7).

$$z_i = f_\theta(x_i) \in \mathbb{R}^k, \tag{7}$$

where the latent dimension k is smaller than the input dimension d ($k \ll d$). With this reduction, the model is capable of accurately representing the most informative features while discarding noise and irrelevant data. The decoder function is represented as in Eq. (8).

$$g_\phi : \mathbb{R}^k \rightarrow \mathbb{R}^d \tag{8}$$

It reconstructs the input by mapping the latent code z_i back to the original space, as mentioned in Eq. (9).

$$\hat{x}_i = g_\phi(z_i) \tag{9}$$

3.2.2. Objective reconstruction

The deep autoencoder is trained to minimize the reconstruction loss, which measures the discrepancy between the original input and its reconstruction across all samples. The objective is to optimize the parameters θ and ϕ to minimize the mean squared error loss, which is defined in Eq. (10).

$$L_{\text{rec}}(\theta, \phi) = \frac{1}{n} \sum_{i=1}^{n} \left\| x_i - g_\phi(f_\theta(x_i)) \right\|_2^2, \tag{10}$$

The features are encoded in the model through loss reduction on the n number of samples. To capture complex interactions, the encoder and decoder employ fully connected layers with nonlinear activations, such as ReLU, which is represented in Eq. (11).

$$h_i^{(l)} = \sigma\left(W^{(l)} h_i^{(l-1)} + b^{(l)}\right), \tag{11}$$

Here, $W^{(l)}$ and $b^{(l)}$ are the weights and bias of the layers, and σ is the non-linear activation function. The decoder inverts inputs from the latent embedding z_i, a representation of the data's underlying manifold $\mathcal{M} \subset \mathbb{R}^d$ in a low-dimensional form.

where,

$$z_i = f_\theta(x_i) \approx \mathcal{M} \tag{12}$$

3.2.3. Performance enhancement with regularization and optimization

Gradient-based algorithms are used to optimize parameters θ and ϕ, and the gradients are calculated using backpropagation (to learn efficiently end-to-end), as shown in Eq. (13).

$$\nabla_{\theta,\phi} L_{\text{rec}} = \frac{2}{n} \sum_{i=1}^{n} \left(g_\phi(f_\theta(x_i)) - x_i\right) \nabla_{\theta,\phi} g_\phi(f_\theta(x_i)). \tag{13}$$

To prevent overfitting, a bottleneck dimensionality k and a loss function with explicit regularization terms, such as weight decay, are employed, enabling robust feature learning and the overall loss function as defined in Eq. (14).

$$L_{\text{total}} = L_{\text{rec}} + \lambda R(\theta, \phi), \tag{14}$$

where R denotes the regularization term (e.g., L_2-norm of weights), and λ controls its influence. The encoder f_θ is trained to generate compact and smooth representations (embeddings, z_i) that serve as effective inputs to downstream modules, including attention mechanisms in transformers.

3.3. Attention-guided feature refinement with transformer encoder

To effectively capture complex dependencies and contextual interactions among features in the compressed latent space, we integrate a Transformer Encoder module to refine the embeddings produced by the deep autoencoder. The input to this module is a matrix of latent embeddings, denoted in Eq. (15).

$$Z = [z_1, z_2, \ldots, z_n] \in \mathbb{R}^{n \times k}, \tag{15}$$

where n is the sample size and k is the decreased feature space. Each vector $z_i \in R^k$ encodes compressed information about the corresponding sample, retaining essential patterns while minimizing noise.

3.3.1. Multi-head self attention

The Transformer Encoder employs a multi-head self-attention mechanism [14], which finds relationships between all features using adaptive weights. It generates queries Q, keys K, and values V via linear transformations, as represented in Eq. (16).

$$Q = ZW_Q, K = ZW_K, V = ZW_V \tag{16}$$

where W_Q, W_K, $WV \in R^{k\times d_k}$ are learned weight matrices, and d_k is the attention head dimension. Attention scores are the scaled dot product of queries and Keys, followed by softmax normalization, as shown in Eq. (17).

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{17}$$

This formulation produces a weighted aggregation of the value vectors, focusing on the most essential features. Multiple attention heads capture diverse aspects of feature relationships, and their outputs are concatenated and linearly transformed.

3.3.2. Normalization and residual connections

The residual connection adds input Z to the attention output, followed by layer normalization, mentioned in Eq. (18).

$$H_1 = LayerNorm(Z + MultiHeadAttention(Z)) \tag{18}$$

This step ensures the original embeddings.

3.3.3. Feed-forward network (FFN)

Subsequently, H_1 passes through a feed-forward network (composed of two fully connected layers with ReLU) for further transformation of the feature representations modelled as in Eq. (19).

$$FFN(H_1) = \sigma(H_1W_1 + b_1)W_2 + b_2 \tag{19}$$

Where W_1, W_2 are learnable weights, b_1, b_2 are biases, and σ denotes the ReLU activation function. A residual connection and layer normalization yield the final embeddings mentioned in Eq. (20).

$$H_2 = LayerNorm(H_1 + FFN(H_1)) \tag{20}$$

3.3.4. Enhanced feature representation

The output $H_2 \in \mathbb{R}^{n\times k}$ provides global, context-aware information, helping the classifier effectively capture feature interactions. This attention-guided refinement enables the informative features while suppressing noise, enhancing the predictive accuracy and robustness of the framework. This approach enables interpretability by allowing for the inspection of attention weights. In summary, the Transformer Encoder block serves as a powerful feature-interaction learner, providing global context for robust high-dimensional classification tasks.

3.4. Ensemble of heterogeneous base classifiers

After the Transformer Encoder refines the latent feature representations, the output is fed into an ensemble of heterogeneous base classifiers — CatBoost, SVM, TabNet, and GMLP.

CatBoost: CatBoost is a gradient boosting framework that constructs an additive sequence of decision trees to minimize a loss $\mathcal{L}$. At each iteration m, the prediction function $F^{(m)}$ is obtained by adding the scaled tree T_m with a learning rate η, as expressed in Eq. (21).

$$F^{(m)}(h_i) = F^{(m-1)}(h_i) + \eta T_m(h_i) \tag{21}$$

Each new tree attempts to fit the residual errors (or pseudo-residuals), which are the negative gradients of the loss function with respect to the current predictions, as shown in Eq. (22).

$$r_i^{(m)} = -\left.\frac{\partial \mathcal{L}(y_i, F(h_i))}{\partial F(h_i)}\right|_{F=F^{(m-1)}} \tag{22}$$

CatBoost's ordered boosting and categorical feature encoding mechanisms help reduce overfitting and bias, making it highly effective for tabular data.

Support Vector Machine (SVM): The SVM classifier finds the optimal hyperplane that maximizes the class margins using the kernel function "k", allowing it to handle non-linearly separable data. It minimizes the regularized hinge loss represented in Eq. (23).

$$\min_{w,b,\xi} \quad \frac{1}{2}\parallel w \parallel^2 + C\sum_{i=1}^{n}\xi_i \tag{23}$$

subject to the margin constraints:

$$y_i\left(w^\top\phi(h_i) + b\right) \geq 1 - \xi_i, \quad \xi_i \geq 0 \tag{24}$$

where $\phi(\cdot)$ is the kernel induced feature mapping, C controls the margin maximization and misclassification, and ξ_i are slack variables allowing soft margins. The decision function is computed as shown in Eq. (25).

$$f(h_i) = \sum_{j=1}^{n}\alpha_j y_j \kappa(h_j, h_i) + b \tag{25}$$

where α_j are dual variables learned during training.

TabNet: TabNet, a deep learning architecture designed for tabular data, applies a sequential attention mechanism that generates feature masks dynamically at each decision step t. The mask M_t is computed by an attentive transformer and selectively focuses on the most relevant features mentioned in Eq. (26).

$$M_t = \sigma(A_t) \tag{26}$$

Where σ denotes a nonlinear activation function (e.g., sparsemax), and A_t is the attention score matrix derived from the input features and preceding steps. The masked feature vector is obtained by element-wise multiplication as represented in Eq. (27).

$$h_i^{(t)} = M_t \odot h_i^{(t-1)} \tag{27}$$

which promotes interpretability by explicitly indicating feature importance and enforces sparsity to reduce overfitting.

Generalized Multi-Layer Perceptron (GMLP): GMLP enhances traditional fully connected networks with gating mechanisms that modulate feature interactions layer-wise. At each layer l, the hidden representation is computed by using Eq. (28).

$$h_i^{(l)} = g\left(W^{(l)}h_i^{(l-1)} + b^{(l)}\right) \odot \sigma\left(V^{(l)}h_i^{(l-1)} + c^{(l)}\right) \tag{28}$$

where $W^{(l)}$ and $V^{(l)}$ are learned weights, $b^{(l)}$ and $c^{(l)}$ are biases, $g(\cdot)$ and $\sigma(\cdot)$ are nonlinear activation functions, and $\odot$ denotes element-wise multiplication. The gating mechanism improves the network's capacity to model complex dependencies. Each base classifier C_j independently produces a probability vector $\hat{y}_i^{(j)} \in \mathbb{R}^c$ over the c classes for instance i, satisfying the simplex constraints shown in Eq. (29).

$$\sum_{m=1}^{c}\hat{y}_{im}^{(j)} = 1, \quad \hat{y}_{im}^{(j)} \in [0,1] \tag{29}$$

Ensemble Aggregation (Weighted): To aggregate the predictions from base classifiers, the ensemble employs a Weighted Soft Voting scheme that linearly combines the outputs, as shown in Eq. (30).

$$\hat{y}_i = \sum_{j=1}^{M}\alpha_j\hat{y}_i^{(j)} \tag{30}$$

Where $M = 4$ is the number of classifiers, and α_j are non-negative weights summing to one and optimized during meta-training or validation. The final class label is the class with the highest combined probability as shown in Eq. (31).

$$\hat{c}_i = \arg\max_{m\in\{1,\ldots,c\}}\hat{y}_{im} \tag{31}$$

This ensemble framework effectively leverages the strengths of tree-based, kernel-based, and neural architectures, enhancing generalization and robustness against various challenges, such as noise, feature heterogeneity, and class imbalance, inherent in real-world tabular data.

3.5. Meta-Level Fusion with Meta-MLP

To combine the strengths of heterogeneous base classifiers, we design a Meta-Level Fusion module that employs a Meta-MLP, a compact feedforward neural network. For each instance, it receives concatenated prediction vectors from all base classifiers to learn data-dependent,

dynamic weights for each classifier. The concatenated input for the ith sample is defined in Eq. (32).

$$y_{\text{concat},i} = \left[\hat{y}_i^{(1)} \parallel \hat{y}_i^{(2)} \parallel \hat{y}_i^{(3)} \parallel \hat{y}_i^{(4)}\right] \in \mathbb{R}^{4c} \tag{32}$$

where $\hat{y}_i^{(j)} \in \mathbb{R}^c$ is the class probability vector from the jth classifier for the ith sample, c denotes the number of classes, and $\parallel$ denotes vector concatenation. The Meta-MLP maps this input to a refined probability distribution as shown in Eq. (33).

$$\hat{y}_i = \text{MetaMLP}(y_{\text{concat},i}) \in \mathbb{R}^c. \tag{33}$$

3.5.1. Construction of Meta-Features

The Meta-MLP consists of fully connected layers with nonlinear activations to model the complex nonlinear interactions among the base classifier predictions. This mechanism facilitates the adaptive, instance-specific weighting schemes that surpass fixed averaging or voting, thereby enhancing overall classification performance by leveraging higher-order correlations. Formally, the Meta-MLP parameters Θ can be expressed as in Eq. (34), yielding its output.

$$\hat{y}_i = \sigma\left(W_L \phi(\cdots \phi(W_1 y_{\text{concat},i} + b_1)\cdots) + b_L\right), \tag{34}$$

Where W_l, b_l are weights and biases of the lth layer, $\phi(\cdot)$ is a nonlinear activation function (e.g., ReLU), and $\sigma(\cdot)$ is the softmax for valid class probabilities. The network's depth and width are design choices that balance model capacity and the risk of overfitting.

3.5.2. Objective training

The Meta-MLP is trained in a supervised manner to minimize the categorical cross-entropy loss over a training set of size n, as shown in Eq. (35).

$$\mathcal{L}_{\text{cls}}(\Theta) = -\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{c} y_{ij} \log \hat{y}_{ij}, \tag{35}$$

Where $y_{ij} \in \{0,1\}$ is the indicator that equals 1 if the actual label of the ith sample is class j, and $\hat{y}_{ij}$ is the predicted probability from the Meta-MLP for class j.

The Meta-MLP effectively leverages complementary information from the base classifiers' outputs, yielding more robust and accurate ensemble predictions. The Meta-MLP's hierarchical structure effectively integrates the benefits of CatBoost, SVM, TabNet, and GMLP, achieving improved classification performance that surpasses fixed-weight ensembles. Algorithm 1 outlines the whole model training process and each phase of the model was summarized in Algorithm 2 to Algorithm 6 available in Appendix A.

Algorithm 1 Trans-ADENet Overall Training Procedure

Require:
- Raw dataset $\mathcal{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ with labels $\mathcal{Y} = \{y_1, \ldots, y_n\}$
- Preprocessed raw data
- Hyperparameters: latent dimension k, batch size B, learning rates $\eta_1, \eta_2, \eta_3, \eta_4$

Ensure: Trained parameters: Autoencoder (θ, ϕ), Transformer Encoder ψ, Base classifiers $\{\varphi_j\}$, Meta-MLP ω
Begin
Stage 1: Train Deep Autoencoder
Initialize autoencoder parameters (θ, ϕ)
for epoch = 1 to Epoch$_1$ **do**
  for each mini-batch $x_b \subset \mathcal{X}$ **do**
    $\mathbf{z}_b \leftarrow \text{Encoder}_\theta(x_b)$ {Latent encoding}
    $x_b^{recon} \leftarrow \text{Decoder}_\phi(\mathbf{z}_b)$ {Reconstruction}
    $\mathcal{L}_{\text{rec}} \leftarrow \text{MSE}(x_b, \hat{x}_b)$
    Update (θ, ϕ) using η_1 to minimize $\mathcal{L}_{\text{rec}}$
  end for
end for
Stage 2: Fine-tune Transformer Encoder
Initialize transformer parameters ψ
for epoch = 1 to Epoch$_2$ **do**
  for each mini-batch $(x_b, y_b) \subset (\mathcal{X}, \mathcal{Y})$ **do**
    $\mathbf{z}_b \leftarrow \text{Encoder}_\theta(x_b)$ {Obtain latent embeddings}
    $\mathbf{h}_b \leftarrow \text{Transformer}_\psi(\mathbf{z}_b)$ {Refine embeddings}
    $y_b^{pred} \leftarrow \text{ClassifierHead}(\mathbf{h}_b)$ {Predict probabilities}
    $L_{\text{cls}} \leftarrow \text{CrossEntropy}(y_b, \hat{y}_b)$
    Update ψ (and optionally θ) using η_2 to minimize L_{cls}
  end for
end for
Stage 3: Train Base Classifiers
Extract latent features $H \leftarrow \text{Transformer}_\psi(\text{Encoder}_\theta(\mathcal{X}))$
for each classifier $C_j \in$ {CatBoost, SVM, TabNet, GMLP} **do**
  Initialize parameters φ_j
  Train C_j on $(H, \mathcal{Y})$ to minimize classification loss
end for
Stage 4: Train Meta-MLP Fusion
Construct meta-training dataset:
for each sample $i = 1$ to n **do**
  Obtain base predictions $\hat{y}_i^{(j)} = C_j(h_i)$ for each classifier j
  Concatenate: $y_i^{\text{meta}} = [\hat{y}_i^{(1)} \| \hat{y}_i^{(2)} \| \hat{y}_i^{(3)} \| \hat{y}_i^{(4)}]$
end for
Train Meta-MLP with inputs y_i^{meta} and labels $\mathcal{Y}$ to minimize classification loss with parameters ω
Return: Trained parameters $(\theta, \phi, \psi, \{\varphi_j\}, \omega)$
End

3.6. Computational complexity

The Trans-ADENet framework is designed with computational efficiency and scalability, with a modular pipeline that supports distributed and parallel training.

3.6.1. Auto-encoder stage

The DAE reduces the original high-dimensional feature space R^d to a much lower-dimensional latent space R^k, where $k \ll d$, reducing computational requirements. The complexity of the encoder and decoder networks scales as $O(dk + k^2)$ per instance. Due to offline pretraining and batched updates, the cost remains manageable.

3.6.2. Transformer Encoder stage

The Transformer Encoder processes the reduced embedding matrix $Z \in \mathbb{R}^{n\times k}$ through multi-head self-attention, which has a theoretical time and space complexity of $\mathcal{O}(nk^2)$. Since k is significantly smaller than the original input dimensionality d, and the sequence length n corresponds to the batch size (not the entire dataset), the attention computation remains tractable. Each attention head computes as shown in Eq. (17). The total attention cost per layer remains $\mathcal{O}(nk^2)$, which is scalable for low-dimensional embeddings.

3.6.3. Base classifiers

The base ensemble classifiers — CatBoost, SVM, TabNet, and GMLP — are trained independently and in parallel; the slowest learner bounds the overall training time, not the cumulative cost. The SVM, with its kernel complexity, scales as $\mathcal{O}(n^2 k)$ in the worst case. However, since it is trained on already reduced and refined embeddings, the runtime remains acceptable. CatBoost and TabNet offer GPU-accelerated training, while GMLP remains lightweight due to its shallow architecture.

3.6.4. Fusion layer (Meta-MLP)

The Meta-MLP fusion layer, which aggregates outputs from all base classifiers, operates on concatenated vectors of size $4c$ (where c is the number of output classes). Its complexity scales linearly with the number of ensemble members and is given by Eq. (36).

$$\mathcal{O}(4c \cdot h_m) \tag{36}$$

Where h_m is the hidden layer size of the Meta-MLP. Since this layer is shallow and trained on a small fused representation, it adds minimal overhead to the total cost.

Table 1
Computational complexity of the models.

Model	Complexity
AutoEncoder	$O(dk + k^2)$
Transformer Encoder	$O(nk^2)$
CatBoost	$\approx O(T \cdot n \log n)$
SVM (RBF)	$O(n^2 k)$
TabNet	$O(nd_{\text{tab}})$
GMLP	$O(nkh)$
Meta-MLP Fusion	$O(4c \cdot h_m)$
Trans-ADENet	$O(dk + k^2 + nk^2 + T_{\text{ensemble}} + T_{\text{meta}})$

Table 2
Dataset description.

Dataset name	Ref	Records	Feature size	Classes
Breast GSE45827	[57]	151	54 676	6
Leukaemia GSE9476	[57]	64	22 284	5
INCART 2-lead Arrhythmia	[58]	175 729	34	5
Thyroid RNA	[59]	572	20 530	2

3.6.5. Overall complexity

Overall, the Trans-ADENet complexity is primarily driven by the attention mechanism, resulting in an overall time complexity of approximately that shown in Eq. (37).

$$\mathcal{O}(dk + k^2 + nk^2 + T_{\text{ensemble}} + T_{\text{meta}}) \tag{37}$$

Where T_{ensemble} and T_{meta} denote the training costs of the base classifiers and the Meta-MLP, respectively. The compact latent embeddings and parallel structure ensure that Trans-ADENet scales efficiently with high-dimensional datasets and large sample sizes, making it suitable for real-world biomedical classification tasks involving complex and voluminous data, as shown in Table 1.

4. Experimental results

To evaluate the predictive performance, stability, and generalization ability of the developed Trans-ADENet model, Comprehensive experiments were conducted on three datasets, and dataset details are provided in Table 2. Each dataset employed a uniform preprocessing workflow that included missing-value imputation, feature scaling with StandardScaler, and label encoding to ensure consistent model evaluation.

To compare Trans-ADENet with other methods, we evaluated it against five state-of-the-art baseline algorithms: XGBoost, SVM(RBF kernel), TabNet, and a fully connected deep neural network(DNN), along with dropout and batch normalization, a self-attention neural model that can automatically learn high-order feature interactions was added to compare traditional attention mechanisms with the dual-attention approach used by Trans-ADENet.

All models were tested using 5-fold stratified cross-validation with performance metrics precision, macro F1-score, accuracy, recall, and area under the curve (AUC). Trans-ADENet consistently outperformed all baseline models, achieving the best performance on the leukemia and breast datasets, with accuracies of 96.3% and 94.1%, respectively. The enhanced results are due to the combination of autoencoding, dual attention, and ensembling of meta-MLP.

In contrast, the baseline models showed limited ability to generalize across datasets. XGBoost and TabNet achieved comparable performance on the breast dataset; their performance decreased significantly on the arrhythmia dataset due to its temporal complexity. SVM was performed well on the leukaemia dataset due to its clear class separation, but performed poorly on the INCART 2 arrhythmia dataset due to its noise conditions. AutoInt lacks the Transformer Encoder's pair-level refinement, resulting in moderate performance. DNNs are prone to overfitting due to the high feature-to-sample ratio in omics data.

4.1. Performance comparison

To quantitatively assess the effectiveness of the proposed Trans-ADENet, we conducted a comprehensive comparison with five baselines: XGBoost, SVM (RBF kernel), TabNet, DNN, and AutoInt on three datasets. By using identical data preprocessing and gridsearch-based hyperparameters, each model was evaluated using 5-fold stratified cross-validation. Performance was measured using four key metrics: accuracy, macro F1-score, precision, and recall. The results are shown in Table 3.

Trans-ADENet outperformed all baseline models across all three datasets and evaluation metrics, demonstrating improved classification accuracy, class-wise balance (as indicated by macro F1), and generalization. On the Breast GSE45827 dataset, Trans-ADENet achieved 94.1% accuracy, which is 2.8% higher than TabNet. This performance is attributed to the model's ability to denoise redundant genetic features through deep autoencoding and further refine discriminant representations through Transformer-based dual attention. For the Leukemia GSE9476 dataset, Trans-ADENet achieved the highest accuracy of 96.3%, clearly outperforming the best-performing baseline (TabNet, 94.1%). This performance leverages the well-structured, high-dimensional gene expression profiles with dual attention. The INCART 2-lead arrhythmia dataset achieved 92.7% accuracy, outperforming all baselines by more than 3% and highlighting its suitability for physiologically heterogeneous data and the Thyroid RNA dataset also achieving 94.6%, outperforming all baselines. The dynamic meta-ensemble further enhanced the robustness by weighting predictions from baseline models. The results demonstrate the excellence of Trans-ADENets in providing robust and generalizable solutions to high-dimensional biomedical classification tasks. Its consistent performance across domains underscores its practical utility in real-world clinical and omics-based decision-support systems.

4.2. Confusion matrices and ROC curves

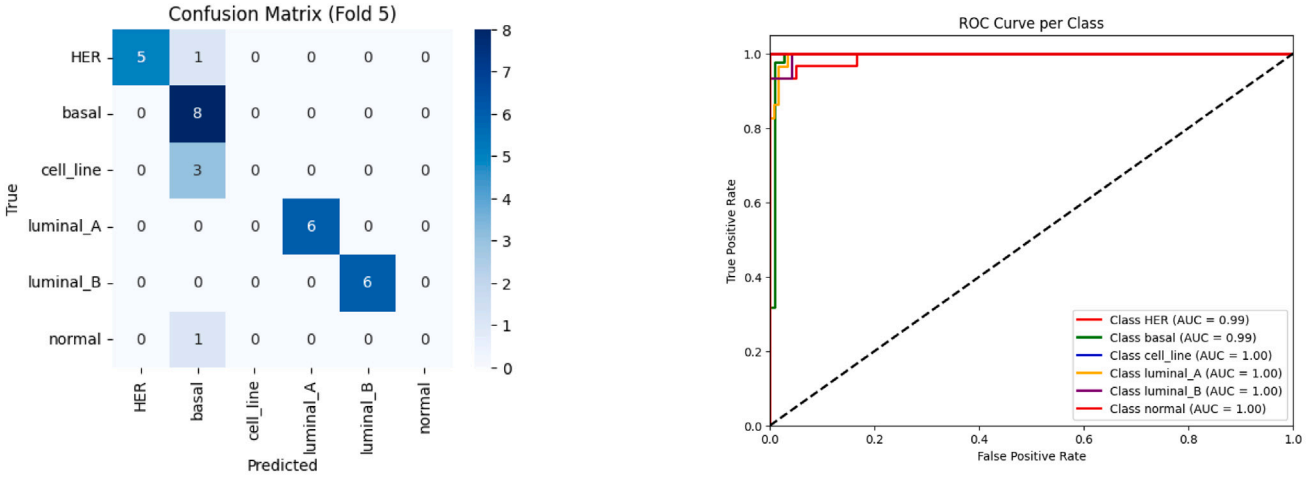
Trans-ADENet's classification performance was evaluated using confusion matrices and ROC curves, which highlighted the sensitivity-specificity trade-off and specific class-specific prediction patterns. The ROC curves and AUC scores characterize discrimination, while the confusion matrices indicate trends in misclassification. This analyses offer complementary insights into the robustness, class-wise balance, and calibration quality of the proposed Trans-ADENet model across diverse biomedical datasets.

The proposed Trans-ADENet was tested on the Breast (GSE45827), Leukemia (GSE9476), and INCART 2-lead Arrhythmia datasets. The results of the fivefold cross-validation and ROC analysis are displayed in Figs. 2 to 4. Each subfigure shows the cross-validation results, along with the associated Receiver Operating Characteristic (ROC) curve, which illustrates the model's threshold-independent discriminative behaviour. The model demonstrated its ability to adjust to different biomedical modalities, including high-dimensional gene-expression and genomic measurements, as well as non-stationary, noisy physiological signals, by achieving AUC values greater than 0.96 with the Breast GSE45827 dataset, greater than 0.97 with the Leukemia GSE9476 dataset, and greater than 0.90 with the INCART 2 lead arrhythmia ECG dataset.

The confusion matrices of the datasets indicate that the predictions are well-balanced, with few false positives and negatives. On the other hand, the ROC characteristics exhibit sharply pointed curves in the upper-left direction, indicating calibrated decision limits and a high sensitivity-specificity ratio. Overall, these results demonstrate that Trans-ADENet's core components — deep autoencoding, transformer-based attention refinement, and meta-ensemble fusion — enable the retrieval of highly discriminative, noise-resistant features and the provision of consistent, interpretable, and generalizable classification results across datasets.

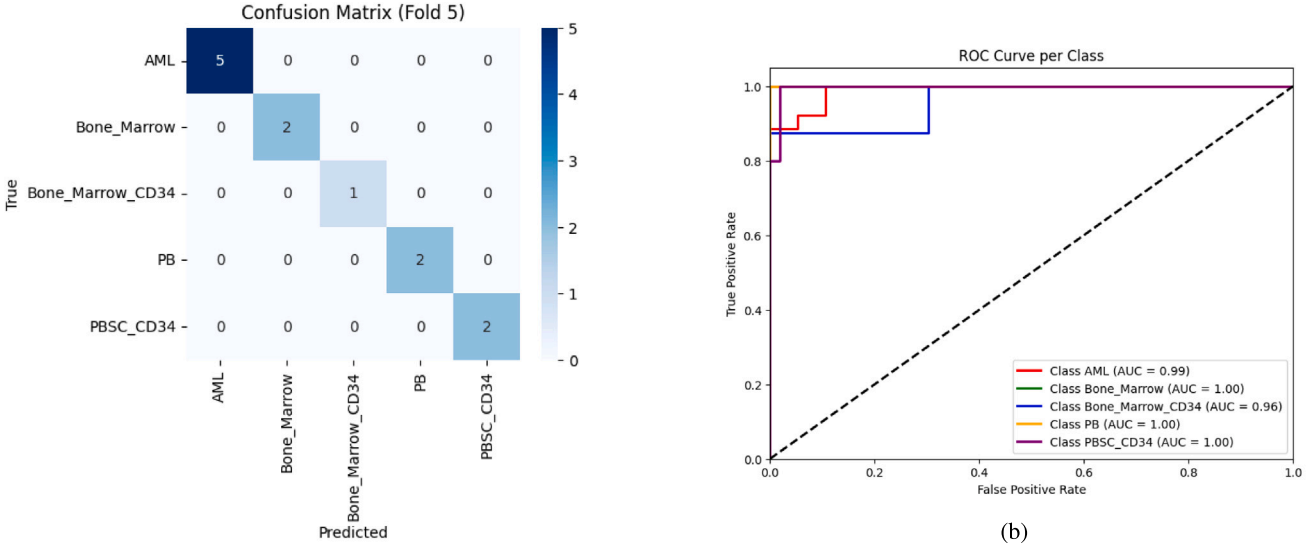
Table 3
Performance comparison across datasets.

Model	Breast_GSE45827 [57]				Leukemia_GSE9476 [57]				INCART 2-lead arrhythmia [58]				Thyroid RNA [59]			
	Acc (%)	F1	Prec	Rec	Acc (%)	F1	Prec	Rec	Acc (%)	F1	Prec	Rec	Acc (%)	F1	Prec	Rec
XGBoost	89.2	0.884	0.888	0.882	92.5	0.923	0.927	0.921	87.6	0.872	0.877	0.869	90.4	0.904	0.863	0.820
SVM (RF)	90.4	0.897	0.901	0.895	93.7	0.936	0.940	0.934	88.2	0.879	0.881	0.878	92.1	0.919	0.889	0.882
TabNet	91.3	0.912	0.916	0.910	94.1	0.942	0.944	0.940	89.3	0.884	0.888	0.882	93.2	0.931	0.907	0.903
DNN	88.7	0.872	0.879	0.868	90.8	0.909	0.912	0.906	85.9	0.861	0.867	0.858	91.7	0.914	0.872	0.869
AutoInt	90.1	0.891	0.894	0.889	92.1	0.918	0.922	0.916	87.5	0.870	0.874	0.869	92.6	0.924	0.889	0.885
Trans-ADENet	94.1	0.938	0.942	0.939	96.3	0.961	0.965	0.963	92.7	0.917	0.921	0.915	94.6	0.950	0.960	0.950



(a) (b)

Fig. 2. (a) Five-fold cross-validation result of Trans-ADENet on the Breast GSE45827 dataset. (b) Mean ROC curve with AUC.



(a) (b)

Fig. 3. (a) Five-fold cross-validation results of Trans-ADENet on the Leukaemia GSE9476 dataset. (b) Mean ROC curve with AUC.

4.3. Ablation study

To rigorously assess the individual contributions of the core components in Trans-ADENet, we conducted ablation experiments on four distinct biomedical datasets: Breast (GSE45827), Leukaemia (GSE9476), INCART 2-lead arrhythmia and Thyroid RNA. The deep autoencoders, Transformer-based dual attention, and meta-MLP ensemble fusion were systematically deleted or replaced. However, the remaining components of the architecture were the same. This Transformer-based dual attention mechanism facilitates contextual feature weighting and a meta-MLP layer designed for ensemble fusion of base classifier outputs. We performed rigorous 5-fold cross-validation to ensure statistical robustness, with accuracy as the key metric. The accuracy indicates that the significance of each component decreases compared to the entire model. The ablation results are shown in Table 4.

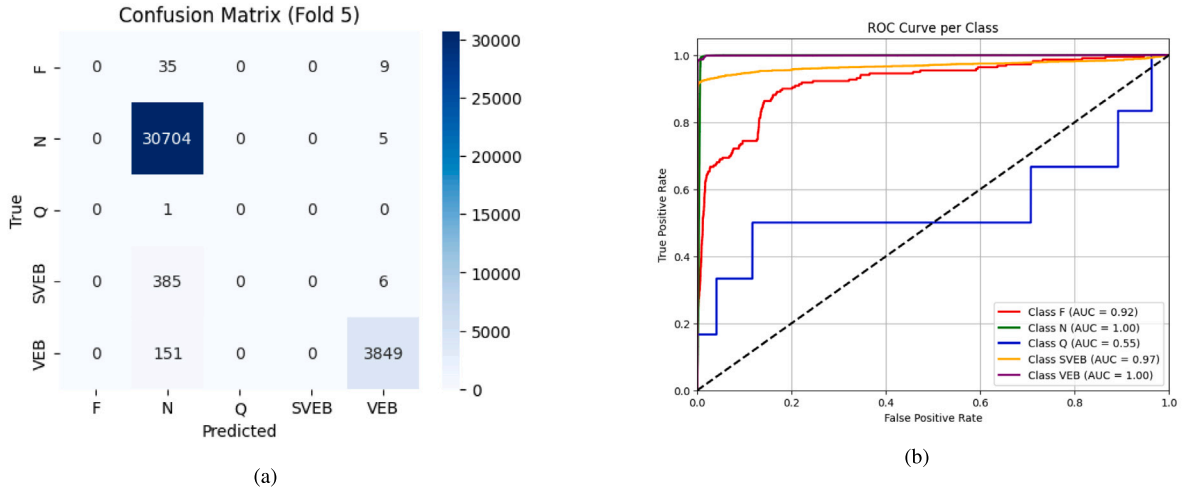


Fig. 4. (a) Five-fold cross-validation results of Trans-ADENet on the INCART 2-lead arrhythmia dataset. (b) The average ROC curve with AUC.

Table 4
Ablation study results across all datasets.

Model variant	Breast accuracy (%)	Leukemia accuracy (%)	Arrhythmia accuracy (%)	Thyroid RNA accuracy (%)
Without Transformer Attention	88.5	91.2	87.0	90.2
Without Autoencoder	84.9	89.3	85.6	91.5
Without Meta-MLP (Soft Voting)	90.2	92.8	89.2	92.0
Proposed (Trans-ADENet)	94.1	96.3	92.7	94.6

The deep autoencoder has the most significant impact on performance. The ablation of this component resulted in the largest drop in accuracy, suggesting that it learns compact, noise-resistant latent representations from the original, high-dimensional, and unstable inputs. This effect is particularly noticeable in the leukemia dataset; omitting the autoencoder resulted in a reduction of approximately 7% in accuracy. This demonstrates that the autoencoder helps to prevent overfitting and improve feature discrimination in high-variance biological data. The Transformer-based dual attention module significantly improves model performance by enabling dynamic, context-aware feature interaction modelling. Without this module, accuracy dropped 5.6% on the Breast dataset and 5.3% on the Arrhythmia dataset. The meta-MLP ensemble fusion layer offered relatively modest performance improvements (a 2%–3% reduction in accuracy upon removal) compared with traditional soft voting strategies. This adaptive fusion improves the model's flexibility and decision-making beyond uniform averaging.

4.4. Interpretability analysis

To enhance model interpretability, a class-wise Shapley Additive Explanation (SHAP) analysis was performed across various datasets, highlighting important discriminative features with biological and clinical significance. Specific groups of high-impact genes were identified in both the Leukemia and Breast Cancer datasets, demonstrating substantial contributions to subtype prediction. In the INCART 2 dataset, significant ECG segments, such as the QRS complex and ST segment, were identified as key clinical markers of arrhythmia.

All datasets and SHAP evaluations for all classes showed consistent, class-specific patterns of feature contributions, suggesting that Trans-ADENet learns physiologically and biologically significant representations. As seen in (Figs. 5 to 9) of the supplementary information (SI), the genes were specifically reduced on the unique wavelengths found in AML and other subtypes in the leukemia dataset. Based on the known molecular heterogeneity among the HER2-enriched, Basal-like, and Luminal subtypes, the results of the SHAP plots (Figs. 10–15) of the breast dataset available in SI, revealed gene significance patterns associated with these subtypes. The model identified waveform signal components that distinguish between ventricular, supraventricular, and normal arrhythmias using the INCART ECG dataset (Figs. 16–20), along with above Thyroid RNA dataset SHAP analysis (Figs. 21–22) also described in the SI. The class-wise SHAP patterns available in SI reveal that Trans-ADENet is based on dataset-specific and discriminative features, ensuring reliable and consistent interpretability across all tasks.

The model interpretation is also assessed with local and global explanations. Local sample-level attention maps show the model considers different latent characteristics to be important for each sample, as expected from the known biological and clinical heterogeneity between patient subgroups. The results of the global SHAP analysis identify a small number of key genes and/or physiological attributes that have class-specific influences on the predictions of the model, many of which correspond to established biomedical markers. This demonstrates that the model's predictive outputs are based upon patterns that are relevant to biology and medicine rather than random associations. All datasets sample level attention(Local Interpretation), SHAP based feature importance(Global interpretation) visualizations (Figs. 23–30) are available in SI.

4.5. Statistical analysis

To evaluate the statistical significance of the performance improvements achieved by Trans-ADENet, we employed the paired Wilcoxon signed-rank test, a nonparametric test used to assess the relationship between two sets of measurements. The test is performed on the same fold for both values (Trans-ADENet vs. baseline model) based on per-fold accuracy scores from 5-fold stratified cross-validation. The Wilcoxon test can be used with continuous, non-Gaussian accuracy values in small-sample biomedical datasets, unlike the paired t-test, which requires a normal distribution of the performance metric. Trans-ADENet was compared to baseline models for each dataset (Breast GSE45827, Leukemia GSE9476, INCART 2 and Thyroid RNA). Under the null hypothesis of equal model effectiveness, the resulting p-values indicate the likelihood that the observed performance differences could

Table 5
Comparing Trans-ADENet vs. baseline Wilcoxon signed-rank test p-values across datasets.

Model	Breast	Leukemia	INCART 2	Thyroid RNA
SVM	0.0212*	0.0012**	0.0318*	0.0225*
CatBoost	0.0215*	0.0239*	0.0412*	0.0281*
TabNet	0.0019**	0.0010**	0.0283*	0.0018**
GMLP	0.0225*	0.0013**	0.0361*	0.0210*
DeepGene	0.0007**	0.0003**	0.0300*	0.0009**
VDTformer	0.0017**	0.0018**	0.0342*	0.0020**

Table 6
Computational cost and training time.

Model	Training time (min)	Avg. epoch time (sec)	GPU memory (GB)
MoVAE	18.4	1.52	1.8
VAE-FastGA	22.1	1.87	2.0
DeepGeneTrans	41.6	3.74	5.2
VDTformer	38.9	3.41	4.8
Trans-ADENet	**24.7**	**2.05**	**2.6**

be due to chance. The symbol "*" shows statistically significant improvement ($p < 0.05$). The symbol "**" indicates statistically highly considerable improvement ($p < 0.01$). The results are shown in Table 5.

4.6. Computational cost and training time analysis

Trans-ADENet's scalability was evaluated by comparing its computational cost and training time with those of baseline models. Table 6 summarizes the total training duration, average epoch time, and GPU memory consumption across all approaches. Despite its complex architecture, which includes an autoencoder, a Transformer Encoder, and a heterogeneous ensemble, Trans-ADENet maintained competitive training costs. The transformer benefits from the autoencoder's compressed latent space, resulting in shorter sequences that reduce GPU memory usage and accelerate convergence compared to other models such as DeepGeneTrans and VDTformer. Additionally, its lightweight ensemble classifiers operate on refined feature vectors, further minimizing computational demands. Trans-ADENet also shows improved convergence compared to MoVAE and VAE-FastGA, thanks to its attention-guided latent refinement. Overall, while DeepGeneTrans provides high accuracy, its high-dimensional data dependence incurs greater costs. Trans-ADENet efficiently processes native biomedical data, achieving a strong balance of accuracy, training time, and computational efficiency, confirming its suitability for large biomedical datasets.

5. Comparison analysis

The proposed Trans-ADENet framework comparison is summarized in Table 7 alongside state-of-the-art methods. In the context of raw high-dimensional biomedical data with large feature-to-sample ratios, Trans-ADENet achieved a high performance of 96.30%, surpassing MoVAE (95.60%) and VAE-FastGA (94.20%). While DeepGeneTrans reports the highest accuracy, its architecture is tailored for biomedical image-like representations and does not natively operate on high-dimensional tabular gene-expression vectors. VDTformer performance (93.90%) is notably lower on these datasets, suggesting that transformer architectures operating directly on full-dimensional biomedical vectors may struggle without an effective dimensionality-reduction mechanism. Trans-ADENet's advantage stems from its integrated design: a deep autoencoder compresses noisy high-dimensional inputs, the Transformer Encoder applies multi-head self-attention to refine latent representations by capturing long-range dependencies, and a heterogeneous ensemble (CatBoost, SVM, TabNet, and GMLP) is fused through an adaptive Meta-MLP. In contrast, VAE-FastGA is not end-to-end trainable and lacks an attention mechanism, limiting its ability to model complex feature interactions. MoVAE employs ensemble learning but lacks adaptive fusion, whereas the Meta-MLP in Trans-ADENet dynamically weights classifier outputs to improve robustness. By contrast, Trans-ADENet is explicitly designed for tabular settings and maintains interpretability while delivering competitive performance.

Table 7
Result comparison.

Approach	Best accuracy	End–End	Attention	Ensemble
MoVAE [60]	95.60%	Yes	Yes	Yes
DeepGeneTrans [54]	97.80%	Yes	Yes	NO
VDTformer [55]	93.90%	Yes	Yes	NO
VAE-FastGA [49]	94.20%	NO	NO	NO
Trans-ADENet (Proposed)	96.30%	Yes	Yes	Yes

6. Conclusion and future work

In this work, we introduce a transformer-enabled attention-guided deep ensemble network, specifically designed to address the unique challenges of classifying high-dimensional biomedical data. The architecture seamlessly integrates three complementary sub-modules: a deep autoencoder for noise removal and dimensionality reduction. This Transformer Encoder captures global contextual feature interactions through self-attention, and a heterogeneous ensemble of classifiers constructed using a meta-level MLP. This integrated framework facilitates consistent and understandable classification, effectively addressing issues such as the curse of dimensionality, feature redundancy, limited sample sizes, and overfitting. Experiments on four different biomedical datasets-breast GSE45827, leukaemia GSE9476, INCART 2 and Thyroid RNA showed that Trans-ADENet outperformed robust baselines in terms of precision, accuracy, recall, F1-score, and AUC. Ablation experiments further confirmed the importance of each module, with significant reductions in performance when any core components were removed. Confusion matrices and ROC curve analysis confirmed the model's robust class discrimination, generalization performance, and stability across folds, even for datasets suffering from noise or class imbalance. There are several exciting avenues for extending this research. First, incorporating multimodal biomedical information, such as fusing genomics, imaging, and clinical data, can improve model performance and usability, particularly by incorporating cross-modal attention mechanisms. Second, applying Trans-ADENet to longitudinal data and temporal relationships will enable its use in disease progression modelling and in real-time patient monitoring. Third, to facilitate deployment in real clinical environments, subsequent releases may explore model compression strategies, such as pruning or knowledge distillation, to enable edge computing or real-time inference. Fourth, incorporating descriptive tools such as SHAP or LIME — broadening the descriptiveness will enable biomarker identification and establish clinical confidence. Finally, validating the model in external cohorts and institutions using domain adaptation or transfer learning will enhance its generalizability and real-world robustness. In summary, Trans-ADENet offers a robust, generalizable, and comprehensible high-dimensional biomedical classification solution with significant potential for future improvement and clinical translation.

CRediT authorship contribution statement

Venkaiah Chowdary Bhimineni: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rajiv Senapati:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Algorithm 2 Preprocessing of Raw Dataset

Require: Raw dataset D with features X and labels y
Ensure: Preprocessed feature matrix Z and encoded label vector $y_{encoded}$
Begin
Load raw dataset D
Remove Features with All Missing Values:
for each feature column j in X **do**
  if all values in column j are missing **then**
    Remove feature column j from X
  end if
end for
Handling Missing Values by Mean Imputation:
for each feature column j in X **do**
  Compute mean μ_j of observed (non-missing) values in column j
  for each missing value x_{ij} in column j **do**
    Replace x_{ij} with μ_j
  end for
end for
Standardize Features:
for each feature column j in X **do**
  Compute mean μ_j and standard deviation σ_j
  for each value x_{ij} in column j **do**
    $z_{ij} \leftarrow \frac{x_{ij} - \mu_j}{\sigma_j}$
  end for
end for
Encode categorical Labels y**:**
Create mapping $LE : \{c_1, c_2, \ldots, c_K\} \rightarrow \{0, 1, \ldots, K-1\}$
for each label y_i in y **do**
  $y_{encoded_i} \leftarrow LE(y_i)$
end for
Return: Preprocessed feature matrix $Z = [z_{ij}]$ and encoded label vector $y_{encoded}$
End

Algorithm 3 Dimensionality Reduction via Deep Autoencoder Training

Require: Dataset $X = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, $\mathbf{x}_i \in \mathbb{R}^d$; latent dimension k ($k \ll d$); learning rate η; number of epochs E; batch size B
Ensure: Encoder parameters θ, decoder parameters ϕ, and latent embeddings $Z = \{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n\}$, $\mathbf{z}_i \in \mathbb{R}^k$
Begin
Initialize encoder parameters θ and decoder parameters ϕ randomly
for epoch = 1 to E **do**
  Shuffle dataset X
  for each batch $\{\mathbf{x}_b\}$ of size B in X **do**
    Encoding:
    for $i = 1$ to B **do**
      $\mathbf{z}_i \leftarrow f_\theta(\mathbf{x}_i)$ (Eq. (7)) {Encoder forward pass}
    end for
    Decoding:
    for $i = 1$ to B **do**
      $\hat{\mathbf{x}}_i \leftarrow g_\phi(\mathbf{z}_i)$ (Eq. (9)) {Decoder forward pass}
    end for
    Compute Reconstruction Loss:
    $L_{\text{rec}} = \frac{1}{B} \sum_{i=1}^{B} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2$
    Compute $\nabla_\theta L_{\text{rec}}$ and $\nabla_\phi L_{\text{rec}}$
    $\theta \leftarrow \theta - \eta \cdot \nabla_\theta L_{\text{rec}}$
    $\phi \leftarrow \phi - \eta \cdot \nabla_\phi L_{\text{rec}}$
  end for
end for
Obtain Latent Embeddings Z **for full dataset:**
for $i = 1$ to n **do**
  $\mathbf{z}_i \leftarrow f_\theta(\mathbf{x}_i)$ (Eq. (12))
end for
Return: Encoder parameters θ, decoder parameters ϕ, latent embeddings Z
End

Algorithm 4 Attention-Guided Feature Refinement with Transformer Encoder

Require: Latent embeddings $Z \in \mathbb{R}^{n \times k}$ (with n samples, k features); projection matrices $W_Q, W_K, W_V \in \mathbb{R}^{k \times d_k}$; output projection $W_O \in \mathbb{R}^{hd_k \times k}$ (h = number of attention heads); Feed-Forward Network (FFN) parameters; Layer Normalization parameters
Ensure: Refined latent representations after Transformer Encoder $H_{\text{refined}} \in \mathbb{R}^{n \times k}$
Begin
Multi-Head Attention:
Initialize empty list $Heads = []$
for each attention head $i = 1, \ldots, h$ **do**
  $Q_i \leftarrow Z \times W_Q[i]$ {Query projections for *head* i}
  $K_i \leftarrow Z \times W_K[i]$ {Key projections *head* i}
  $V_i \leftarrow Z \times W_V[i]$ {Value projections *head* i}
  $\text{Scores}_i \leftarrow \frac{Q_i K_i^T}{\sqrt{d_k}}$ {Scaled dot-product}
  $\text{AttentionWeights}_i \leftarrow \text{softmax}(\text{Scores}_i)$
  $\text{Head}_i \leftarrow \text{AttentionWeights}_i \times V_i$
  Append $Head_i$ to Heads
end for
$\text{ConcatenatedHeads} \leftarrow \text{Concatenate}(\{\text{Head}_1, \ldots, \text{Head}_h\})$
$\text{AttentionOutput} \leftarrow \text{ConcatenatedHeads} \times W_O$ {Final Linear Projection}
Transformer Encoder(Z)**:**
$\text{AttentionOutput} \leftarrow \text{MultiHeadAttention}(Z, W_Q, W_K, W_V, W_O)$
$H_1 \leftarrow \text{LayerNorm}(Z + \text{AttentionOutput})$
$\text{FFN_Output} \leftarrow \text{FeedForwardNetwork}(H_1)$ {FFN:ReLU$(H_1 W_1 + b_1)W_2 + b_2$}
$H_2 \leftarrow \text{LayerNorm}(H_1 + \text{FFN_Output})$
returns H_2
Main refinement process:
$H_{\text{refined}} \leftarrow TransformerEncoder(Z)$
Return: H_{refined}
End

Algorithm 5 Ensemble of Heterogeneous Base Classifiers

Require: Refined latent feature matrix $H = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_n] \in \mathbb{R}^{n \times k}$; labels $Y = [y_1, y_2, \ldots, y_n]$; number of base classifiers $M = 4$; classifier set $C = \{\text{CatBoost, SVM, TabNet, GMLP}\}$; weight vector $\alpha = [\alpha_1, \alpha_2, \alpha_3, \alpha_4]$ with $\alpha_j \geq 0$ and $\sum_{j=1}^{4} \alpha_j = 1$
Ensure: Final predicted class labels $\hat{Y} = [\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n]$
Begin
Initialize empty list Models
Base Classifier Training:
for $j = 1$ to M **do**
  Select classifier C_j from C
  Train classifier C_j on H and Y
  if C_j = CatBoost **then**
    Train gradient boosted trees minimizing loss L
  else if C_j = SVM **then**
    Solve quadratic optimization for maximum margin hyperplane
  else if C_j = TabNet **then**
    Train TabNet with sequential attention masks on H
  else
    Train GMLP with gating mechanisms on H
  end if
  Append trained classifier C_j to Models
end for
Weighted Prediction:
for $i = 1$ to n **do**
  Initialize $\hat{\mathbf{y}}_i \leftarrow \mathbf{0} \in \mathbb{R}^c$ {c = number of classes}
  for $j = 1$ to M **do**
    $\mathbf{p}_i^{(j)} \leftarrow \text{Models}[j].\text{predict_proba}(\mathbf{h}_i)$
    $\hat{\mathbf{y}}_i \leftarrow \hat{\mathbf{y}}_i + \alpha_j \cdot \mathbf{p}_i^{(j)}$
  end for
end for
Final Class Assignment:
for $i = 1$ to n **do**
  $\hat{y}_i^{\text{class}} \leftarrow \arg\max_{m \in \{1, \ldots, c\}} \hat{\mathbf{y}}_i[m]$
end for
Return: $\hat{Y} = [\hat{y}_1^{\text{class}}, \ldots, \hat{y}_n^{\text{class}}]$
End

Algorithm 6 Training Meta-MLP for Meta-Level Fusion

Require: Base classifier predictions for each sample i: $\hat{\mathbf{y}}_i^{(1)}, \hat{\mathbf{y}}_i^{(2)}, \hat{\mathbf{y}}_i^{(3)}, \hat{\mathbf{y}}_i^{(4)} \in \mathbb{R}^c$, where c is the number of classes, and $\hat{\mathbf{y}}_i^{(j)}$ is the predicted class probabilities from base classifier j; Ground truth labels: $\mathbf{y}_i \in \{0,1\}^c$ (one-hot encoded); Meta-MLP parameters $\{\mathbf{W}_l, \mathbf{b}_l\}$ with activation $\phi(\cdot)$ and softmax $\sigma(\cdot)$; Learning rate α, epochs E, batch size B.
Ensure: Trained Meta-MLP parameters Θ and fused predictions $\hat{\mathbf{y}}_i$ for each sample i
Initialize Meta-MLP parameters $\Theta = \{\mathbf{W}_1, \mathbf{b}_1, \ldots, \mathbf{W}_L, \mathbf{b}_L\}$ randomly
for $e = 1$ to E **do**
  Shuffle training data
  for each mini-batch of samples $i_1, i_2, - - - - i_B$ **do**
    for each sample i in mini-batch **do**
      Construct feature vector: $\mathbf{y}_{\text{concat},i} = \hat{\mathbf{y}}_i^{(1)} \| \hat{\mathbf{y}}_i^{(2)} \| \hat{\mathbf{y}}_i^{(3)} \| \hat{\mathbf{y}}_i^{(4)} \in \mathbb{R}^{4c}$ (Eq.(32))
    end for
    for each sample i in mini-batch **do**
      Forward pass through Meta-MLP: $\mathbf{z}_1 = \phi(\mathbf{W}_1 \mathbf{y}_{\text{concat},i} + \mathbf{b}_1)$; $\mathbf{z}_2 = \phi(\mathbf{W}_2 \mathbf{z}_1 + \mathbf{b}_2)$; $\ldots$; $\mathbf{z}_L = \mathbf{W}_L \mathbf{z}_{L-1} + \mathbf{b}_L$; $\hat{\mathbf{y}}_i^{\text{meta}} = \sigma(\mathbf{z}_L)$
    end for
    Compute categorical cross-entropy loss over mini-batch: $\mathcal{L}_{\text{cls}} = -\frac{1}{B} \sum_{i \in \text{batch}} \sum_{j=1}^{c} y_{ij} \log \hat{y}_{ij}^{\text{meta}}$
    Back propagate loss and compute gradients $\nabla_\Theta \mathcal{L}_{\text{cls}}$
    Update Meta-MLP parameters $\Theta \leftarrow \Theta - \alpha \nabla_\Theta \mathcal{L}_{\text{cls}}$
  end for
end for
for each sample i **do**
  Construct $\mathbf{y}_{\text{concat},i} = \hat{\mathbf{y}}_i^{(1)}, \hat{\mathbf{y}}_i^{(2)}, \hat{\mathbf{y}}_i^{(3)}, \hat{\mathbf{y}}_i^{(4)}$ (Eq.(32)) and compute $\hat{\mathbf{y}}_i = \text{MetaMLP}(\mathbf{y}_{\text{concat},i})$ (Eq.(33))
end for
Return: Trained Meta-MLP model Θ and predictions $\hat{\mathbf{y}}_i$

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.array.2025.100659.

Data availability

All the dataset used for this research work is cited in the manuscript.

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